

Saikumar Mallarapu

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PROFESSIONAL SUMMARY

Python Django Developer with 1 year of experience designing and shipping Django-based production systems across healthcare, NGO, and education domains. Delivered 5+ applications with REST APIs, normalized PostgreSQL schemas, role-based access control, and query-optimized ORM layers — focused on backend correctness, clean API architecture, and scalable data modeling.

WORK EXPERIENCE

Python Django Developer

July 2025 – Present

M7 Corporation

Nellore, India

- Designed and shipped 5+ Django applications across healthcare, education, and NGO domains — including relational database schema design, REST API development using Django REST Framework, and server deployment via Unicorn + Nginx on Linux-based environments
- Built RESTful API response structures using DRF Serializers with nested relationships, field-level validation, and custom error handling — reducing frontend integration issues across multiple client projects
- Developed a Bulk Email Automation system using Django and SMTP, eliminating 60% of repetitive manual dispatch work by replacing a fully manual process with a scheduled, list-driven workflow
- Designed a multi-client Invoice Generator with dynamic template rendering, per-user branding, and PDF export via WeasyPrint — supporting reusable invoice schemas across different client configurations
- Built a barcode-based receipt management system that auto-generates unique transaction codes, replacing manual receipt tracking and enabling instant record lookup by scan — deployed across 2 foundation client projects
- Performed structured API testing using Postman (auth flows, request validation, error response checks) and wrote unit tests for critical Django views and model logic using Python's unittest framework to validate core business workflows

PROJECTS

Bhaarath Medical College & Hospital Management Portal | *Django, DRF, PostgreSQL, JavaScript, Bootstrap*

- Designed a normalized PostgreSQL schema with indexed foreign key fields across doctor, service, staff, and media entities — enabling efficient multi-table filtering queries on the admin dashboard
- Built 10+ RESTful API endpoints using DRF ViewSets and Router-based URL configuration with token authentication — consumed by a JavaScript frontend for dynamic content rendering without page reloads
- Implemented role-based access control (Admin, Staff, Doctor) using Django's permission framework — scoping data access and dashboard visibility per role without duplicating view logic
- Developed session-based authentication with login throttling to protect against brute-force access on sensitive hospital data entry points
- Conducted 3 on-site client requirement sessions, translated business workflows into data models and API contracts, and delivered the full system within a 6-week timeline

GCC Workforce Education & Park Management System | *Django, PostgreSQL, Excel Integration, Bootstrap*

- Architected a multi-module dashboard with hierarchical filtering (zone → centre → batch) — optimized cross-table queries using Django ORM `select_related` and `prefetch_related` to reduce redundant database hits across staff, student, and attendance datasets
- Built staff approval verification workflows with status transitions (Pending → Approved → Rejected), enforcing role-based action permissions across education and park management modules
- Integrated Excel export for attendance and student records using openpyxl — enabling non-technical administrators to access structured data without database access
- Implemented multi-field search and filter across large datasets with indexed query fields, improving query performance and reducing lookup time for admin users managing daily operations

Invoice Generator & Business Automation Suite | *Django, WeasyPrint, SMTP, Python*

- Built a multi-user Invoice Generator with per-account template customization, dynamic field binding, and PDF export via WeasyPrint — structured to support isolated invoice schemas per registered client
- Developed a Bulk Email Bot with SMTP integration that processes recipient lists and dispatches templated campaign emails in batch — replacing a fully manual send process

- Created a Label Bot for internal business process automation, reducing repetitive data entry operations across routine workflows

Personal Project: Task Manager REST API | *Django REST Framework, PostgreSQL, JWT Auth, Swagger*

- Built a production-structured REST API with JWT-based authentication (access + refresh token flow) and role-based endpoint access using DRF permission classes
- Applied Django ORM query optimization using `select_related` and `prefetch_related` on list endpoints to minimize redundant database hits across relational data
- Documented all API endpoints using drf-spectacular (OpenAPI / Swagger), generating an auto-updated interactive API schema for client consumption
- Structured codebase into modular Django apps (`users/`, `tasks/`, `api/`) with `requirements.txt`, `.env.example`, and README covering setup, architecture, and endpoint reference

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL

Frameworks: Django, Django REST Framework, Bootstrap

Databases: PostgreSQL, MySQL, SQLite, Django ORM, Query Optimization, Database Indexing

APIs & Auth: REST API Design, JWT Authentication, Token Auth, Session Auth, OTP Auth, DRF Serializers, DRF ViewSets, Swagger / OpenAPI

Tools: Git, GitHub, VS Code, WeasyPrint, openpyxl, SMTP, Linux / Ubuntu, Gunicorn, Nginx

Concepts: MVC Architecture, Role-Based Access Control, Relational Schema Design, API Response Structuring, Async Email Dispatch, PDF Generation, Barcode Systems, Unit Testing (unittest)

Tools & Platforms: Docker, Redis, Celery, AWS EC2 / S3, Pytest, GitHub Actions CI/CD, API Testing (Postman)

EDUCATION

Master of Computer Applications (MCA)

2025 – Present

Shree Krishna Chaitanya Institute of Science & Technology

Nellore, India

Bachelor of Computer Applications (BCA) – Data Science

2022 – 2025

Shree Krishna Chaitanya Degree College

Nellore, India

CERTIFICATIONS

Certified Web Developer (Upskilling) | NIELIT | June 2025 | Grade: B

LANGUAGES

Telugu (Native) · English (Professional) · Tamil (Conversational) · Hindi (Basic)